NVIDIA GTX 590 and XFX Radeon HD6990

The war for supremacy rages on

Gaming enthusiasts are always on the lookout for the flagship graphics card of every generation. Last generation, we had seen the balance tilting in favour of AMD which was aggressively coming out with DX 11 ready HD 5xxx series of cards.

Lately NVIDIA has more than covered lost ground, thanks to its GTX 5xx series of cards which are performing the HD 6xxx series cards. Till AMD announced its latest generation dual GPU flagship card - the HD 6990 - the only other fastest dual GPU card was its own ie. Radeon HD 5970. NVIDIA's GTX 295 was no match for the HD 5970 as was seen across benchmarks. But this time around, within two weeks of AMD's HD 6990 launch, NVIDIA announced the GTX 590, its dual-GPU answer to HD 6990. We got a chance to pit off these cards against each other. But before moving on to the performance charts, let's get some dope on the innards of the cards.

NVIDIA GTX 590 is basically made up of two GTX 580s having the GF110 architecture. So in a way multiply the specs of GTX 580 by 2 and you have the GTX 590. The clock speeds though have been reduced to 607 MHz (core clock) and 3414 MHz data rate (memory clock) to keep the temperatures under check. The card is 11-inches long and has two 8-pin power connectors.

NVIDIA GTX 590

The XFX Radeon HD 6990 is based on two Cayman XT GPUs. The Cayman GPU was previously seen on HD 6970. So the HD6990 is two HD 6970s and has twice the transistors, shader processors and raster operators. The HD 6990 has a BIOS switch just beside the CrossfireX connector which allows you to operate the card in factory mode (core clock: 830 MHz) in position 1 and in the unlocked mode (core clock: 880MHz) in position 2. The HD 6990 is comparitively longer at 12-inches and has two 8 - pin power connectors.

Both the cards are back-plated and have a centrally located fan section with vapour chamber based cooling for the two GPUs. Both cards come with a multi-GPU bridge.

XFX Radeon HD 6990

We ran both the cards on their factory settings ie. 607 MHz for GTX 590 and 830 MHz for HD 6990. The table below explains how the cards fared against each other.

Based on performance, the NVIDIA GTX 590 egdes past the HD 6990 in all our test games except ***Crysis Warhead***, surprisingly. But the difference is not very prominent in most of the games except ***STALKER: Call of Pripyat***. Pricing of the GTX 590 at ₹44,250, makes us recommend it over the HD 6990 which comes at ₹48,000.

Although we know that for such cards, the pricing makes very little difference. It all boils down to what games you will be playing and at what resolutions. Above 60 fps everything is comfortably playable and with both these cards, that is a given - at the most extreme of game settings.

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Cards	**XFX HD6990**	**NVIDIA GTX 590**
Brand	XFX	NVIDIA
Model	HD 6990	GTX 590
Price (₹)	48000	44250
Score (out of 100)	**84.66**	**88.34**
Type of memory / Memory Clock (MHz)	GDDR5 / 5000	GDDR5 / 3414
Core Clock (MHz)	830	607
No. of Stream Processors	3072	1024
Memory size (MB) / Memory Interface	4096 / 512	3072 / 768
Connects on card	1xDVI, 4xmini DisplayPort	3xDVI, 1x mini
Synthetic Benchmarks		
3D Mark 11 (Overall)	8476	8264
Unigine Heaven v2.1 (1920x1080, 4xAF, Max Shader, Extreme Tesellation) Score	1615	1840
Unigine Heaven v2.1 (1920x1080, 4xAF, Max Shader, Extreme Tesellation) Frame rate	64.1	73
Temperature (Load)	84	81
Game Benchmarks		
FarCry 2 (1680x1050, Ultra High, 8xAA) Avg	132.45	133.35
FarCry 2 (1920x1080, Ultra High, 8xAA) Avg	130.94	131.22
S.T.A.L.K.E.R : Call of Pripyat (Day, 1680x1050, 4xAA, DX11, Ultra Detail) Avg	125.6	142.5
S.T.A.L.K.E.R : Call of Pripyat (Day, 1920x1080, 4xAA, DX11, Ultra Detail) Avg	116.1	134.3
Resident Evil (1680x1050, 8xAA, Motion Blur on)	116.6	124.6
Resident Evil (1920x1080, 8xAA, Motion Blur on)	110.9	120
Crysis Warhead (1920x1080, DX10, Very High) Avg	56	52
Crysis Warhead (1920x1080, DX10, Very High With Tweaks) Avg	55	51
Ratings		
Features	8.5	8
Performance	8	8.5
Build Quality	8	8
Value for money	7	7
Overall	8	8